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MVP OSM

a tool that allows to individuate the areas of high activity based on level of detail

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SoNet Group– http://sonet.fbk.eu



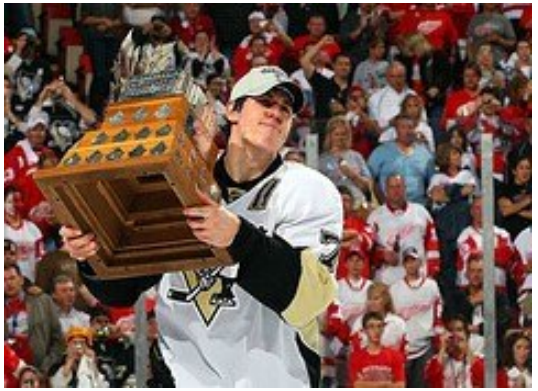
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OpenStreetMap like a game



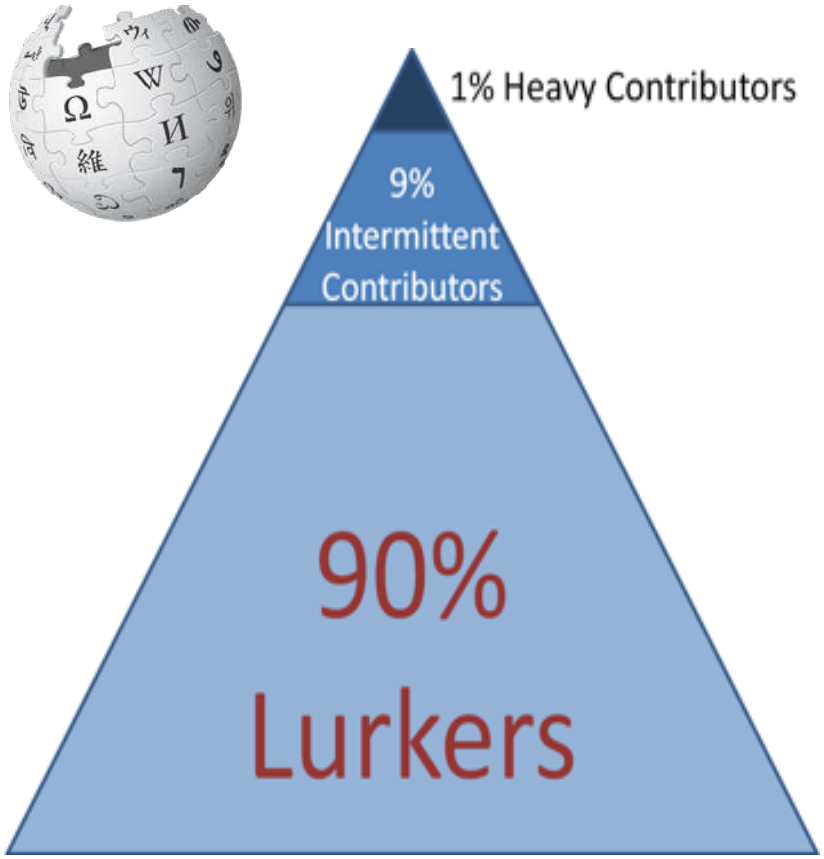
Who are the best players?

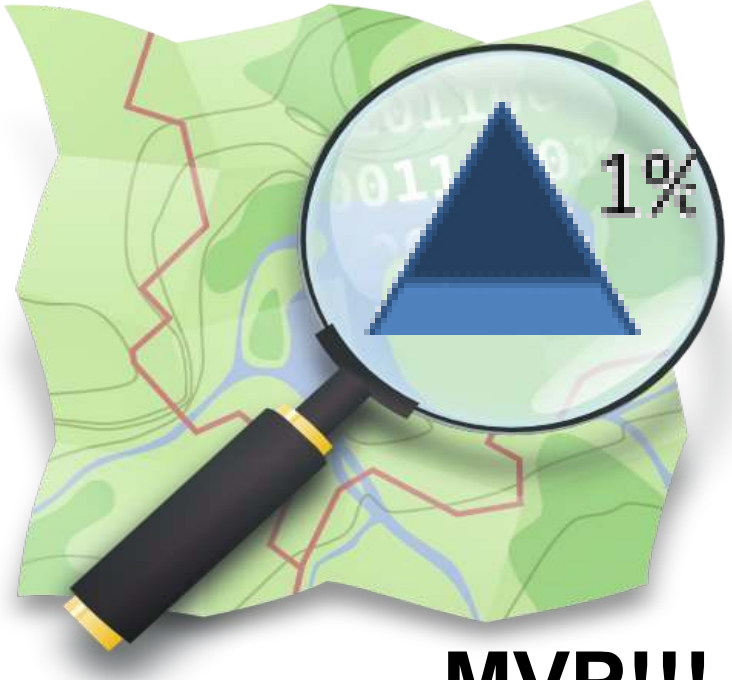
Who are the **M**ost **V**aluable **P**layers?



Images from:http://www.flickr.com/photos/sully_aka__wstera2

The 90-9-1 Principle





MVP!!!

Jakob Nielsen's Alertbox, October 9, 2006 inShare68
Participation Inequality: Encouraging More Users to Contribute

http://www.useit.com/alertbox/participation_inequality.html 1-9-90

How find?

Common Sense:
a user is a good player if gives many contributions to the map

a test with an italian user



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Nodes created by simone



? ?
? ?

The complete Italy???

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… he is also a “data” importer …



[…]
I'm the prime guilty for the import of
the italian internal borders between
municipalities, provinces, regions, and country
[…]

http://wiki.openstreetmap.org/wiki/User:Simone#Data

Solution

Restrict the query only to some details, those requiring

Local knowledge
Experience
Recognition

A good example:



`step_count=*`

44 keys selected

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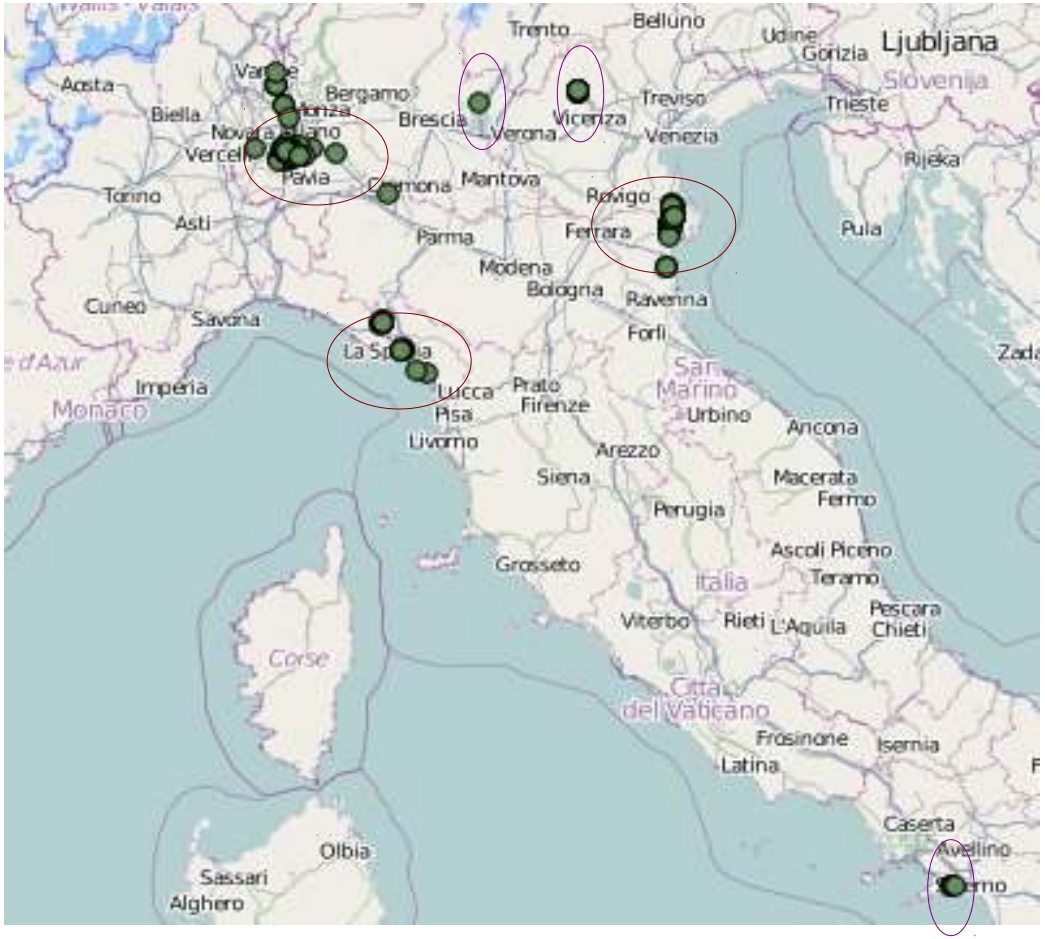
Keys ...

abandoned
access
access:bicycle
access:bus
access:foot
access:hgv:max_length
access:motorcar
amenity
bicycle
bicycle:backward
dispensing
disused
drinkable
embankment
foot
foot:backward
footway
hiking
horse
incline
oneway

recycling:batteries
recycling:cans
recycling:clothes
recycling:engine_oil
recycling:glass
recycling:paper
recycling:plastic
recycling:plastic_bottles
recycling:plastic_packaging
recycling:scrap_metal
recycling:white_goods
recycling:wood
step.condition
step_count
step.height
step.length
surface
surface.material
traffic_calming
traffic_sign
trail_visibility
visibility

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this is the new result



Locations:

Personal life (family, vacancy, work)

OSM life (mapping party, short journeys ...)

The “pet location” concept

have a great care for a location as well as a lovable pet





http://matt.dev.openstreetmap.org/owl_viewer/

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Pet location vs Mapping Party

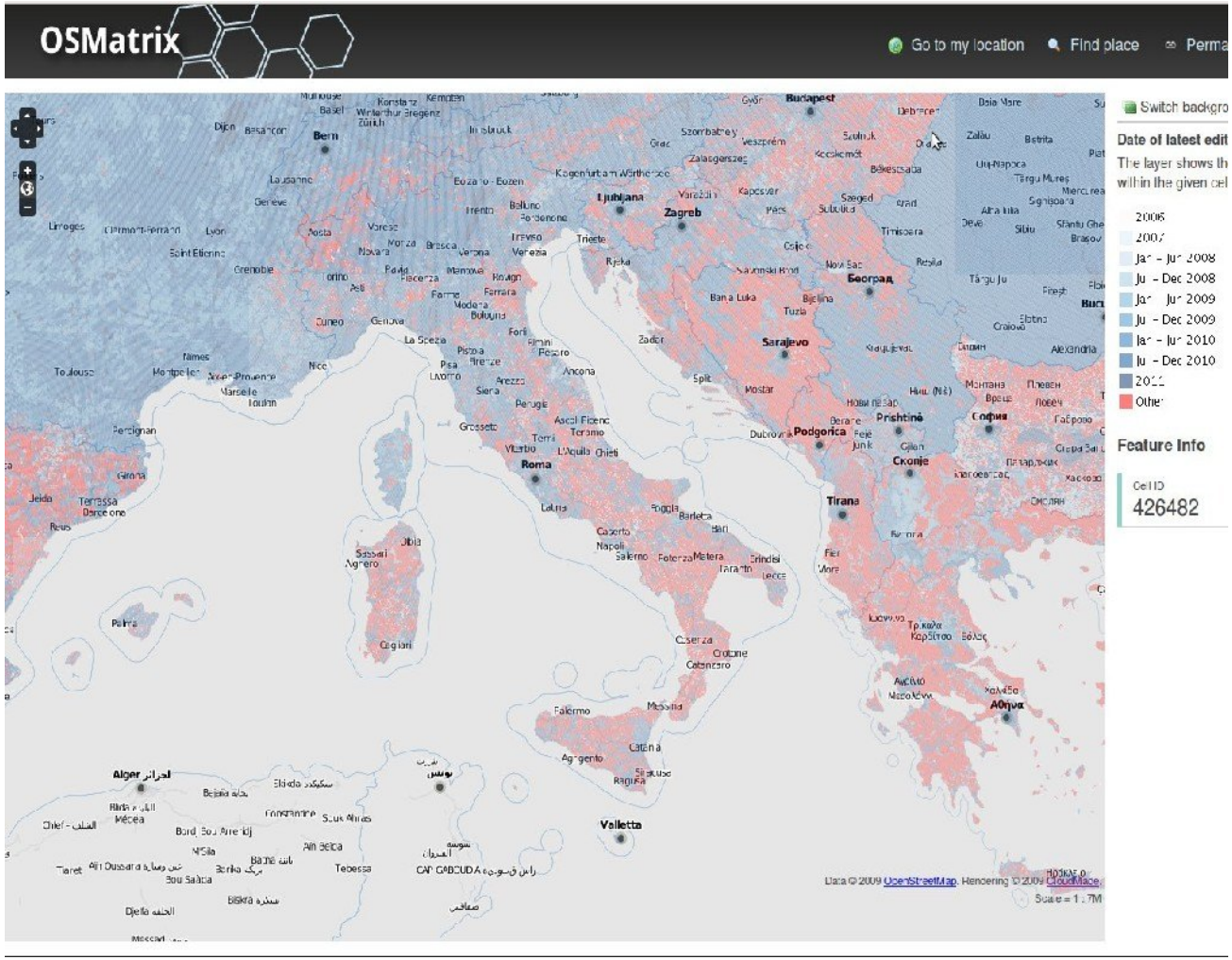
You will always find a mailbox that isn't mapped

Edoardo Marascalchi
italian osm mapper

frequency of update

...but there is always a little bit of noise

Like in OSMatrix



Date of last edit

Linus Law
"given enough eyeballs, all bugs are shallow"
Eric Raymond

http://koenigstuhl.geog.uni-heidelberg.de/osmatrix/

Crowd Quality concept

attempts to quantify the 'collective intelligence of the crowd generating data' in a spatio-temporal context.

User quality

Local knowledge,
Experience and Recognition

→ Like a MVP player

Feature quality

How many different users contributed to a feature?
How has a feature developed over time?

"The impact of crowdsourcing on spatial data quality indicators"
M. van Exel, E. Dias, S. Fruijtier - 2010

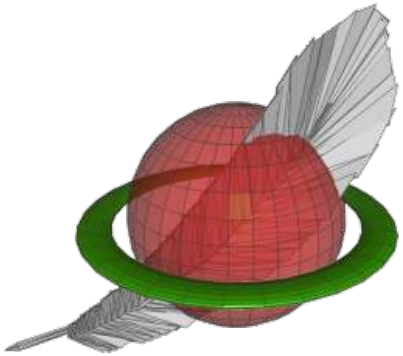
How MVP OSM works

osm xml

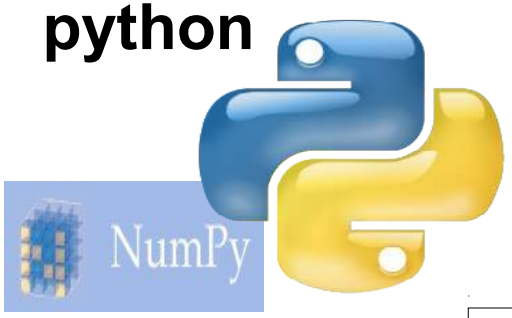
```
<?xml version='1.0' encoding='UTF-8'?>
<osm version="0.6" generator="Osmosis 0.32">
  <bound box="45.68083,10.37794,47.09520,12.48232" origin="http://
www.openstreetmap.org/api/0.6"/>
  <node id="9191867" version="4" timestamp="2010-09-01T22:36:26Z"
uid="330007" user="pikappa79" changeset="5658159" lat="46.104119"
lon="11.0899633"/>
  <node id="9191891" version="5" timestamp="2009-03-11T13:30:43Z"
uid="104478" user="giacos73" changeset="786213" lat="46.3693138"
lon="11.290448"/>
  <node id="9191920" version="3" timestamp="2009-03-10T12:43:40Z"
uid="104478" user="giacos73" changeset="775964" lat="46.2262194"
lon="11.1514362"/>
```

spatialite_osm_**raw**

spatialite



python



sql

differet vectors

for a gis analysis

https://github.com/napo/mvp-osm

The imported tables structure

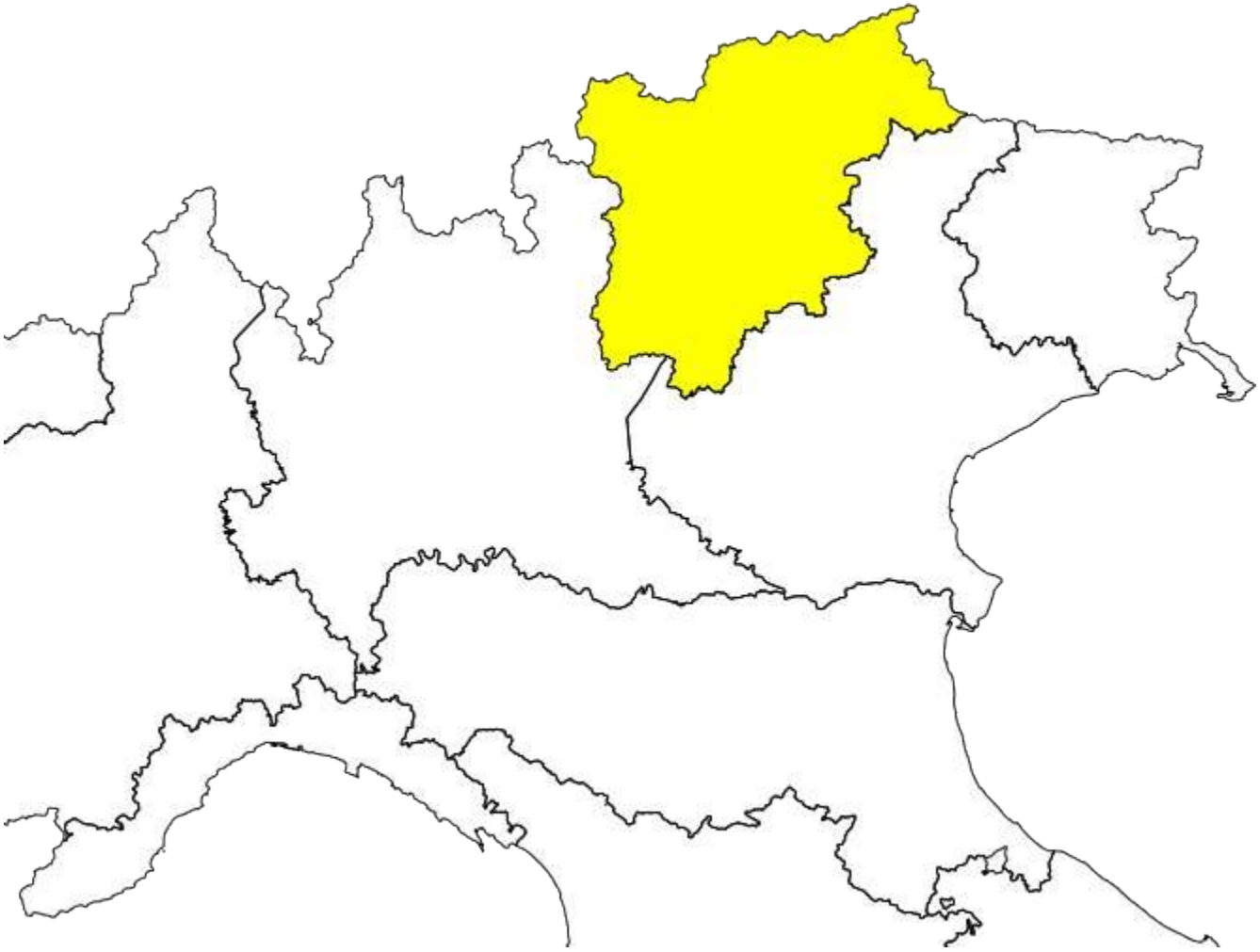
```
spatialite_osm_raw -o map.osm -d database.sqlite
```

```
osm_relation_way_refs
osm_relations
osm_way_node_refs
osm_node_tags
osm_way_tags
osm_nodes
osm_ways
osm_relation_node_refs
osm_relation_relation_refs
osm_relation_tags
[...]
```


GEOMETRY (points)

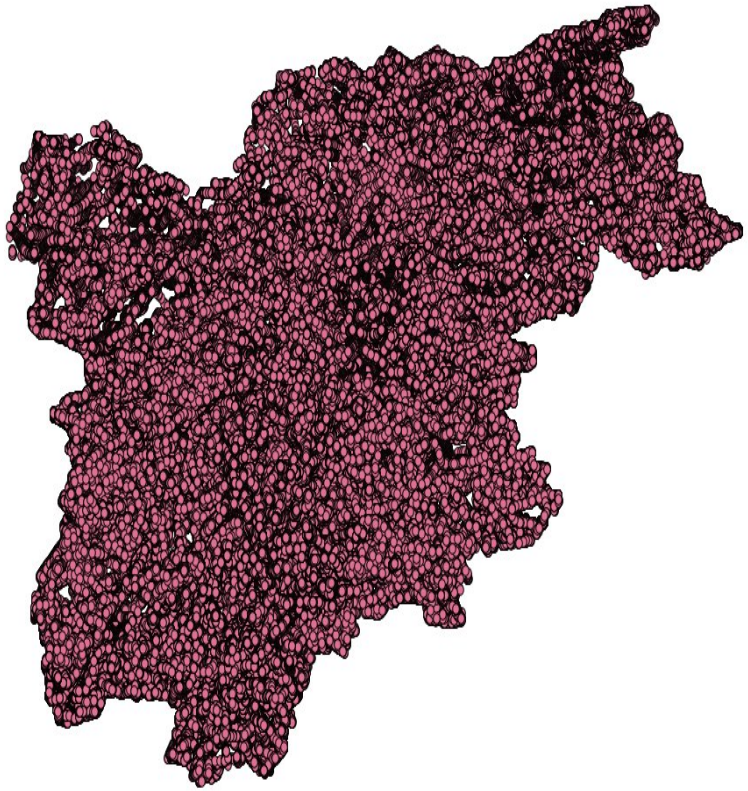
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An example with Trentino Alto Adige



STEP 1/3 – find the points

Query to extract the details



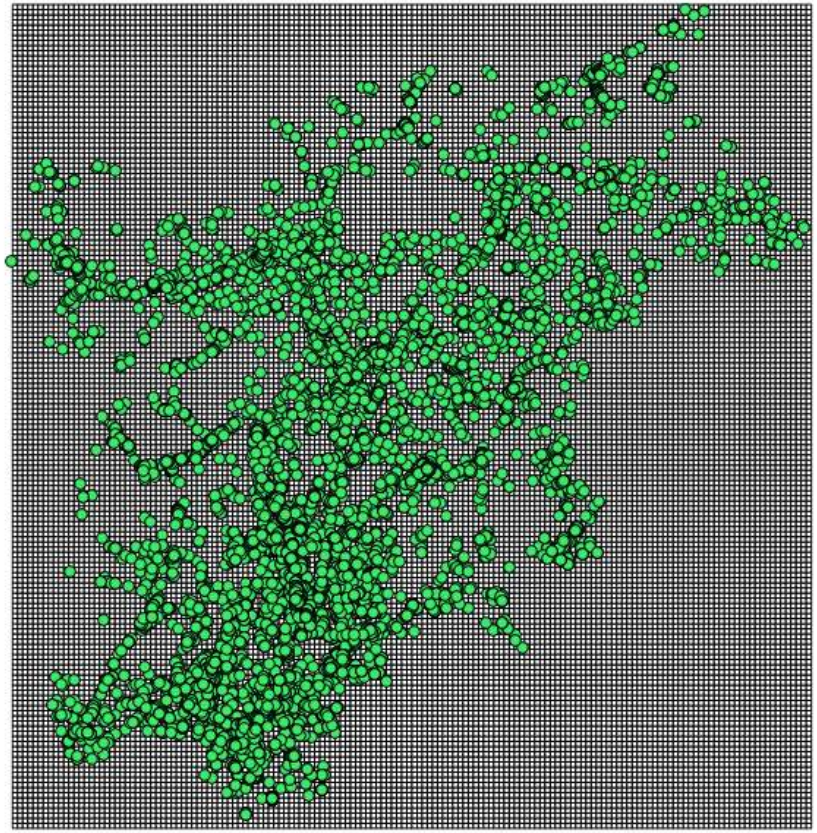
1253 users contributed to this map.
2465872 points

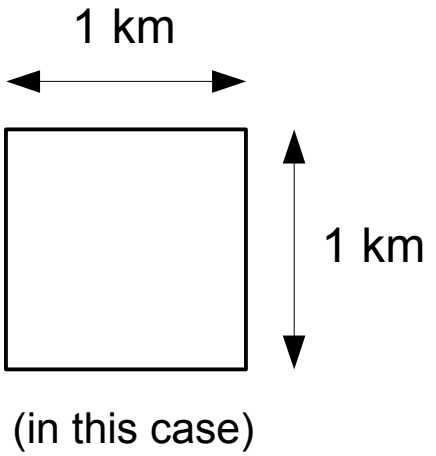


600 users match the query
13675 points

STEP 2/3 - Density

create a grid ...



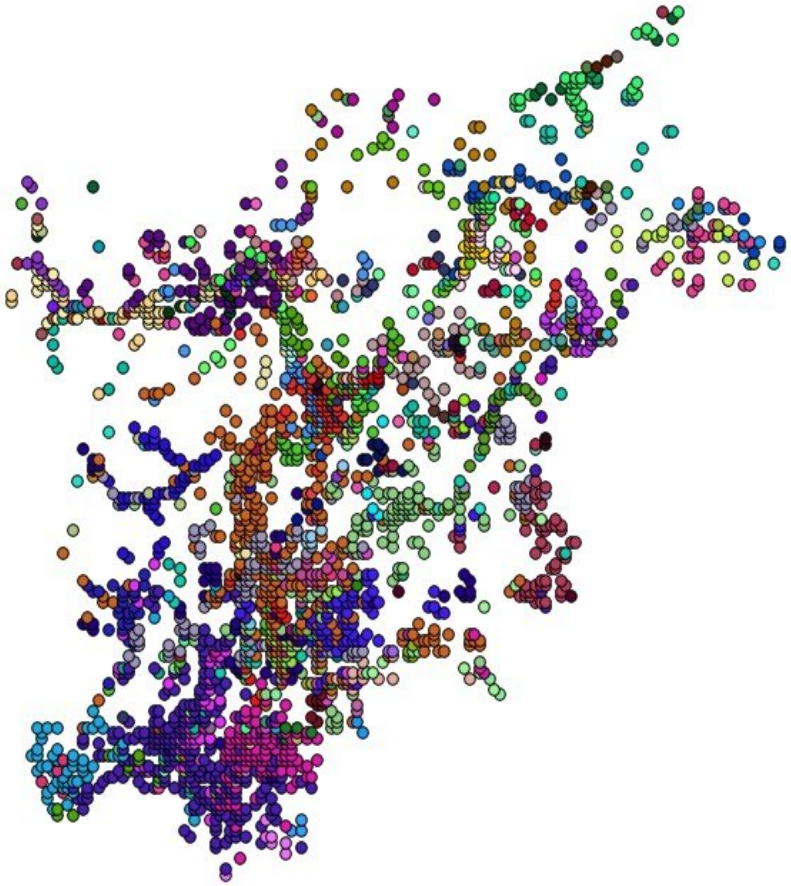


(in this case)

STEP 2/3 - Density

... calculate the data density for each user

2.444 cells



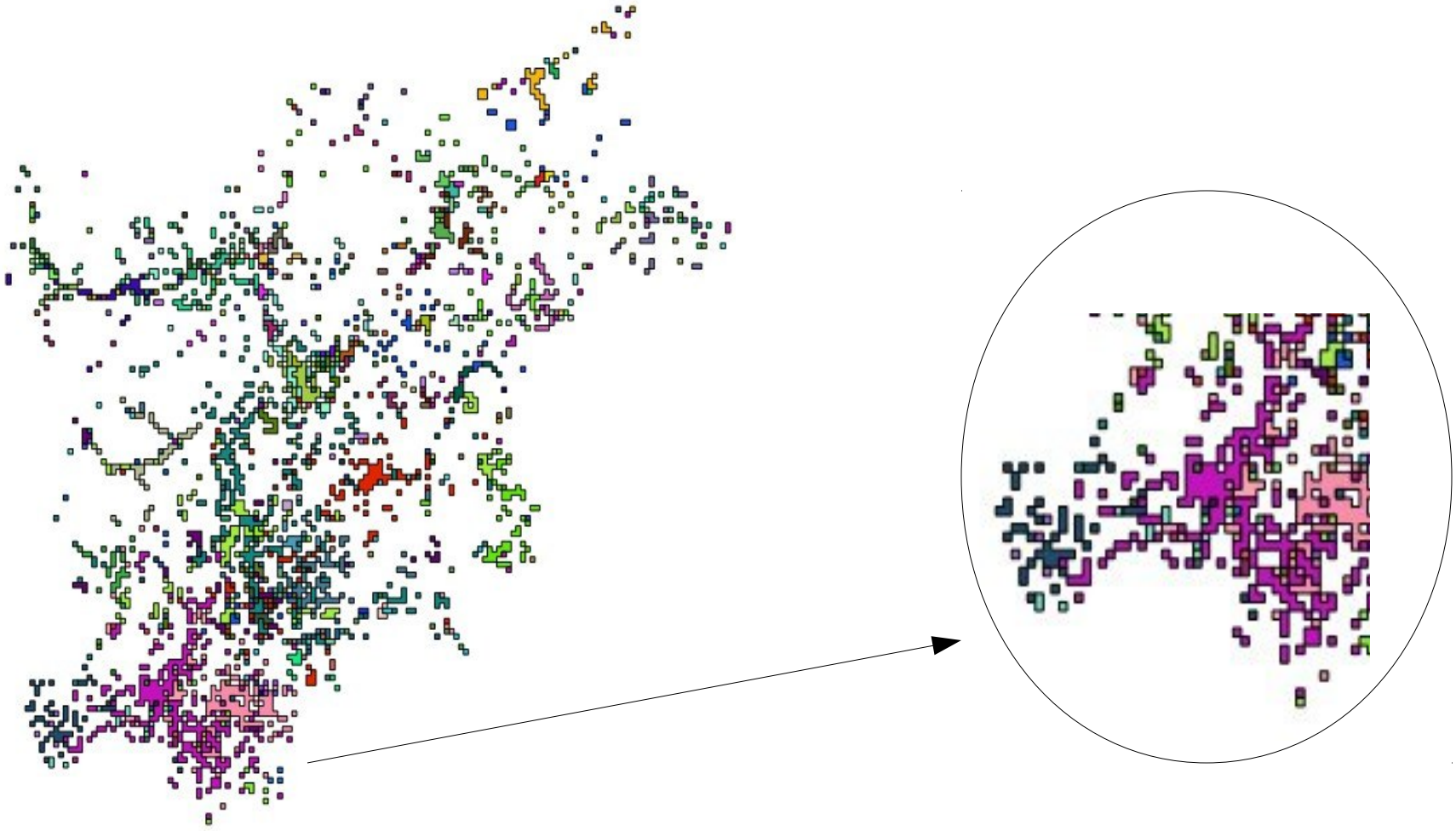
- Al3xius
- AlbA
- Alexander Roalter
- AndiVille
- Andre68
- Andreas Cierpka
- Axel2009
- Bepo
- BergFrido
- BogNik
- Claudius Henrichs
- ColinMarquardt
- Donsen64
- Eckloff
- Emos62
- Fabry

this operation calculate only the users active over the past 3 months

In this case 147 users

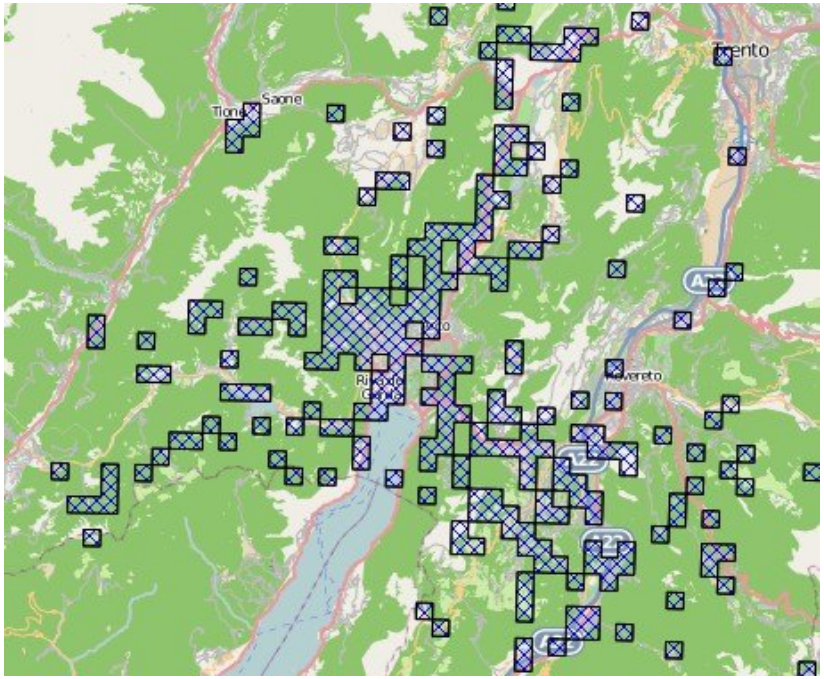
STEP 3/3 - cluster

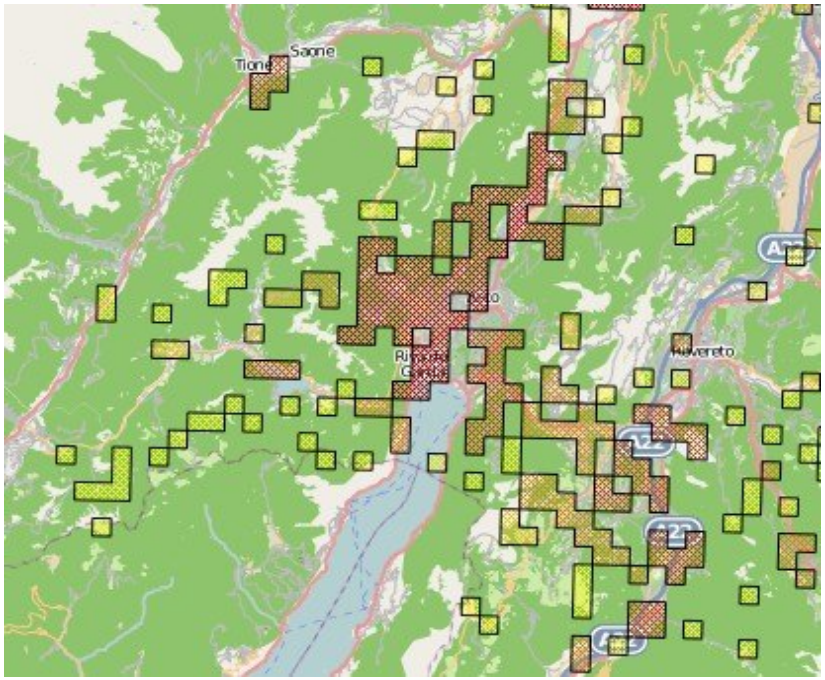
cluster the data for each user by distance



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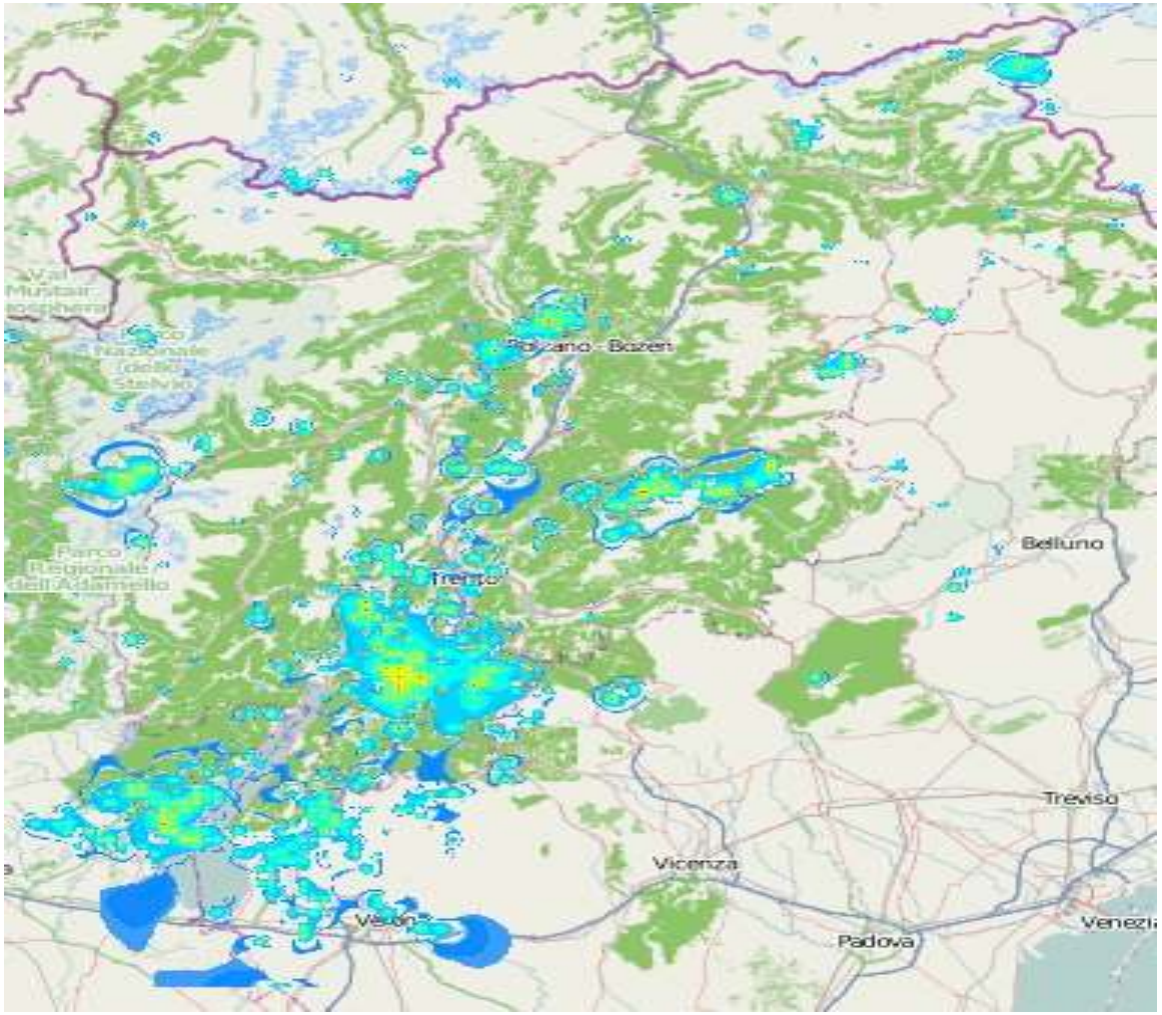
Data Analisys (1/2)





Pet location for a user (1) and where is the most activity (2)

Data analisys 2/2



heat map of the locations

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End

Thanks for the attention

Special thanks to Alessandro Furieri
(spatialite head developer)
for the support



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